Research on Machine Learning Methods for Predicting Risk of Cardiovascular Disease

I. Introduction

This study aims to explore the value of applying machine learning techniques in cardiovascular disease risk prediction [1], by building and comparing multiple classification models to identify the optimal prediction scheme. The core rationale for this methodology is that cardiovascular disease diagnosis is essentially a typical two-class problem: determining whether a patient has cardiovascular disease risk based on their clinical characteristics. Traditional risk assessment methods often rely on doctors' empirical judgments or simple scoring systems, whereas machine learning methods can automatically learn complex nonlinear relationships from multidimensional clinical data, thereby providing more accurate and objective predictions.

In the model selection strategy, this study adopts a methodological framework of multi-model comparisons. The necessity of this choice is reflected in the following aspects: First, different machine learning algorithms have their own theoretical assumptions and applications, such as logistic regression hypotheses that there are linear relationships between features and logarithmic probabilities, while decision trees and integration methods can capture more complex nonlinear patterns; Second, by systematically comparing multiple models, different algorithms can reveal differences in performance when processing medical data, providing empirical grounds for model selection in real applications; Finally, multi-model comparisons also help identify intrinsic characteristics of datasets, such as the interaction effects between features and the degree of nonlinear relationships. This study aims to obtain deep insights into aspects such as model performance, feature importance, and hyperparameter sensitivity through this comparison process, thereby providing scientific guidance for the actual deployment of cardiovascular disease prediction systems. The GitHub link is https://github.com/songbo327/disease-analysis.

II. Problem statement

Cardiovascular disease is one of the main causes of death globally [2]. According to the World Health Organization, about 17.9 million people die from cardiovascular disease annually, accounting for 31% of total global deaths [3]. In this severe context, accurate early risk assessments of cardiovascular disease are crucial for preventive medical interventions and the development of personalized treatment programs. This study defines the cardiovascular disease prediction problem as a standard binary task that determines whether a patient has cardiovascular disease based on multidimensional clinical characteristics. Specifically, given a patient record that includes characteristics such as age, gender, height, weight, blood pressure, cholesterol levels, blood sugar levels, and lifestyle factors (such as smoking, alcohol consumption, physical activity), the model needs to output a binary label indicating whether the patient has cardiovascular disease risk.

The importance of this issue is reflected not only in its enormous public health value but also in its close association with clinical practice. From an academic research perspective, cardiovascular disease prediction spans multiple interdisciplinary fields, including epidemiology, biostatistics, and computational medicine, and the findings can provide theoretical support for identifying disease risk factors and developing intervention strategies. From an industry application perspective, accurate prediction models can be integrated into electronic health record systems, assisting doctors in clinical decision-making and achieving the transition from passive treatment to active prevention. From a practical impact perspective, early identification of high-risk populations can significantly reduce medical costs, improve patient quality of life, and reduce the overall burden of the healthcare system. Therefore, this study has not only theoretical value but also significant social and practical significance.

III. Data Collection and Preprocessing

The data used in this study is derived from the publicly available cardiovascular disease dataset on the Kaggle Platform (sulianova/cardiovascular-disease-dataset), which was originally compiled from patient medical examination records collected by medical institutions and contains rich clinical characteristics. The publicity and widespread use of the dataset ensure the reproducibility of the study and the comparability of the results. The original dataset contained 70,000 samples, each with 13 attributes, including 1 identifier, 11 characteristic variables, and 1 target variable. The characteristic variables can be divided into three categories: objective fact-class characteristics, including age, sex, height, and weight; medical examination-result-class characteristics, including systolic and diastolic blood pressure, cholesterol, and blood sugar levels; and subjective information-class characteristics, including smoking status, alcohol consumption status, and physical activity levels. The target variable,

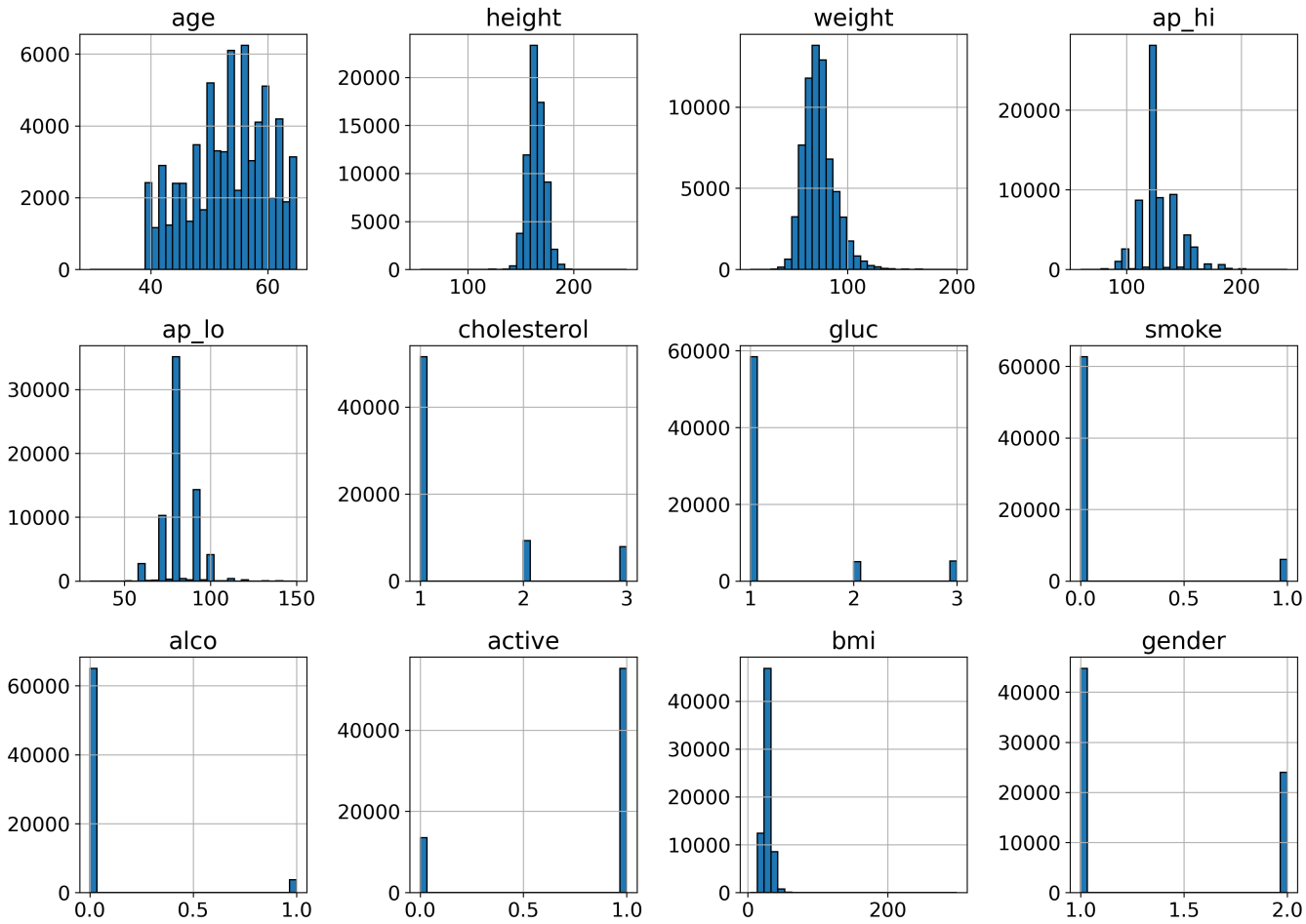

Fig. 1. Feature distributions of the cardiovascular disease dataset.

cardio, is a binary indicator of whether the patient has been diagnosed with cardiovascular disease (Figure 1).

A descriptive statistical analysis of the dataset revealed several important characteristics (Figure 2). In terms of sample number, 68,737 valid samples were ultimately retained after pre-processing, and the data volume was sufficient to support reliable model training and evaluation. In terms of characteristic distribution, the age variable was presented in years after conversion, with an average of 53.29 years and a standard deviation of 6.76 years, indicating that the sample was mainly concentrated in the middle-to-old age group; the distribution of height and weight variables was relatively reasonable, with averages of 164.36 centimeters and 74.12 kilograms, respectively; the systolic average was 126.62 mmHg and the diastolic average was 81.35 mmHg in the blood pressure variable, which was in the normal high range. In terms of category balance, the distribution of target variables presented good balance characteristics, with a disease-free sample of 34,721 (50.5%), a disease-affected sample of 34,016 (49.5%), and a category ratio of 1.02, this balanced distribution avoiding potential interference of category imbalance problems in model training.

The data pre-processing phase employs systematic cleansing and conversion strategies to ensure data quality and model training effectiveness [4]. First, we remove the identifier column, which does not contribute to the prediction task. Second, we convert the age variable from days to the standard years format, improving feature interpretability. Third, we calculate body mass index (BMI) from height and weight, an important derived feature for assessing obesity. The formula for calculating the BMI is body weight (kg) divided by height (m) squared. Fourth, we performed duplicate-record detection and deletion to ensure the dataset's uniqueness and integrity. Fifth, we implemented outlier filtering for blood pressure variables, maintaining reasonable ranges for systolic pressure between 50 and 250 mmHg and diastolic pressure between 30 and 150 mmHg, thereby eliminating potential measurement or data-entry errors. Together, these preprocessing steps ensure that subsequent modeling work is based on high-quality, high-reliability

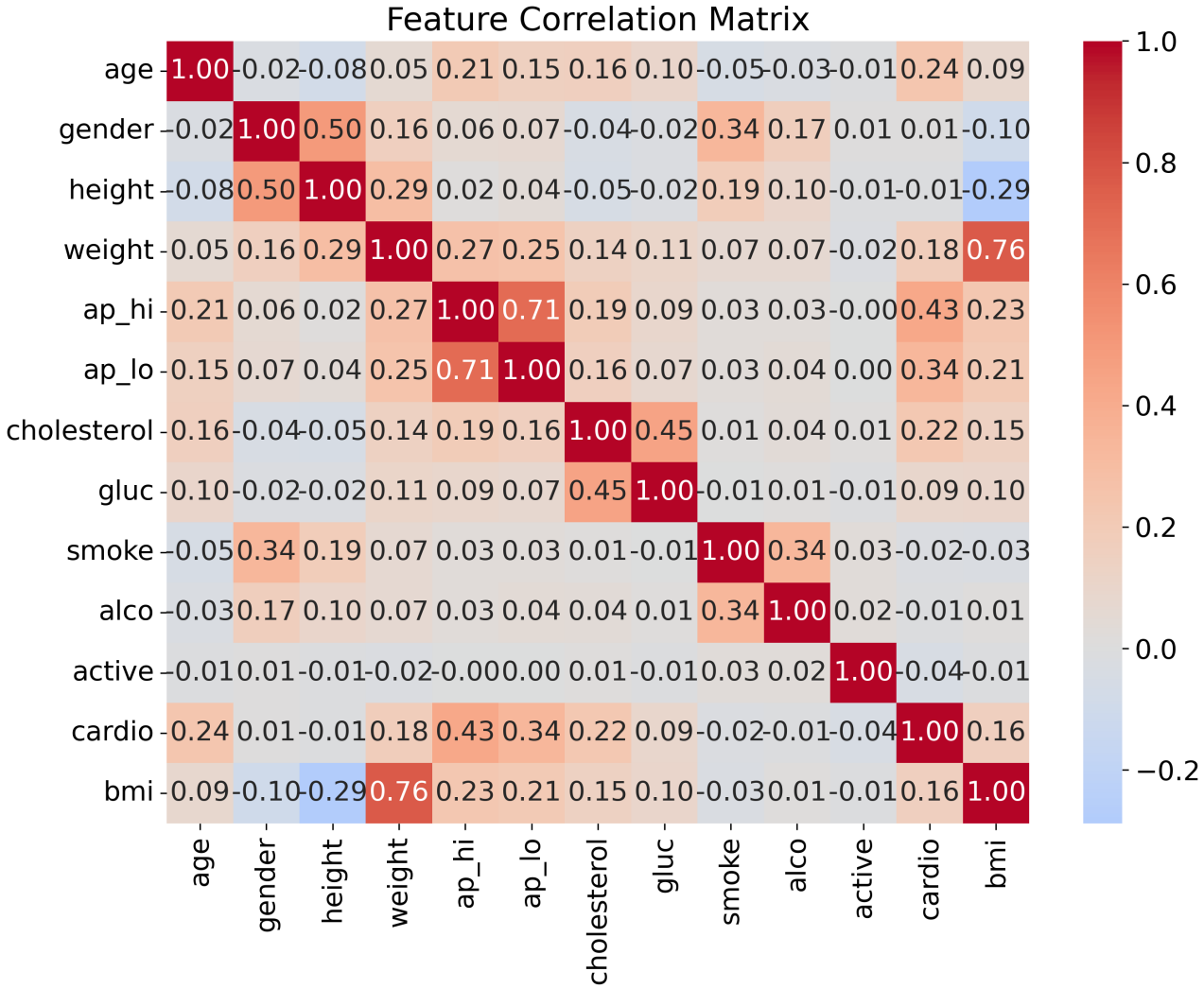

Fig. 2. Correlation matrix of features.

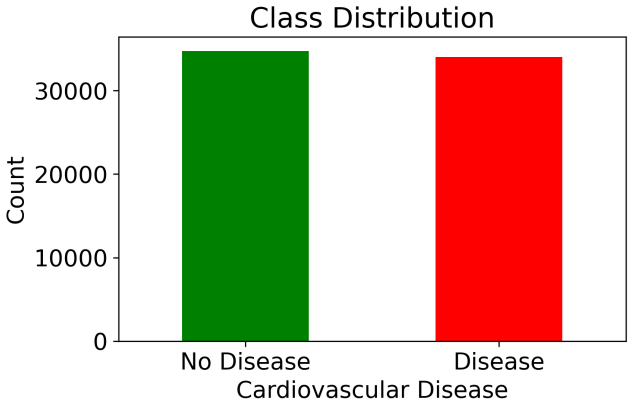

Fig. 3. Class distribution of the cardiovascular disease dataset.

data.

IV. Machine Learning Models Selection

In the model selection process, this study considered multiple factors, including the model's theoretical characteristics, computational complexity, interpretability, and applicability to medical data. After careful evaluation, five representative machine learning models were ultimately selected for comparative study, spanning the spectrum from linear to nonlinear, from single to integrated models [5].

As a classic linear classification model, logistic regression has been widely studied for its strong theoretical foundation, high computational efficiency, and easy-to-explain results [6]. Logistic regression assumes a linear relationship between features and the log of the probability, which is simple yet performs well in many practical applications, especially when the relationship is approximately linear.

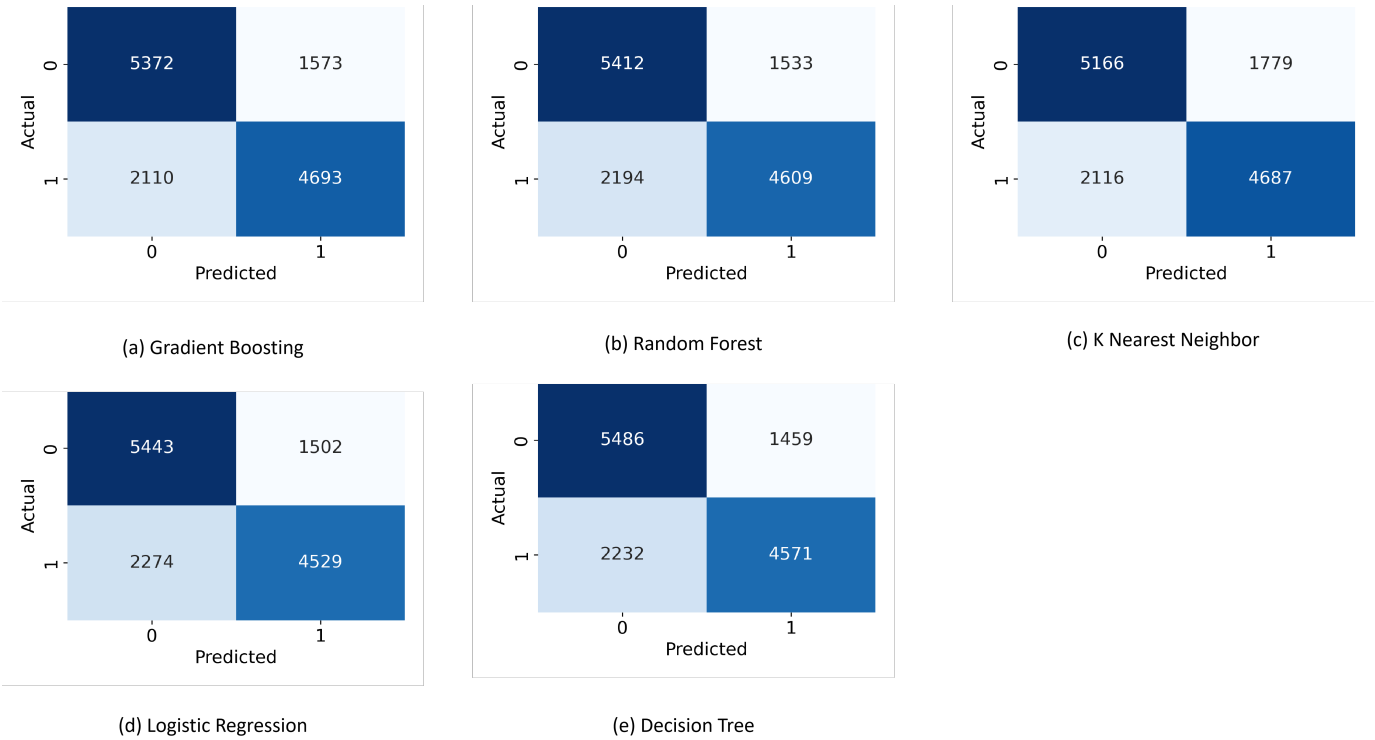

Fig. 4. Confusion matrix of models.

Moreover, coefficients in logistic regression can be directly interpreted as the degree to which features affect predictions, a feature that is particularly important in medical applications, as doctors need to understand the model's decision-making basis. K-nearest neighbors classifiers are chosen as an instance-based learning method because of their nonparametric nature: they do not make strong assumptions about the data distribution and can flexibly adapt to diverse data patterns. The K-nearest neighbors algorithm classifies by computing distances between test samples and training samples, based on the core idea that similar samples should have similar labels.

Decision trees, as rule-based models, are widely studied because they can automatically discover interaction effects and nonlinear relationships among features, while the generated decision rules are highly interpretable and easy for clinicians to understand and accept. Random forests [7], as an integration extension of decision trees, effectively reduce the risk of overfitting from a single decision tree by building multiple decision trees and synthesizing their prediction results, improving the model‘s ability to generalize. Random forests can also provide feature importance estimates, helping identify the features that contribute most to predictive performance. Gradient boosting trees [8], another powerful integration method that iteratively trains weak learners to gradually reduce prediction errors, show excellent performance in numerous machine learning competitions and practical applications. The advantage of gradient boost trees is that they can automatically handle feature interactions, are more robust to outliers, and can balance model bias and variance by adjusting the learning rate and tree depth.

The primary reason for comparing more models in this study is that different models may exhibit significant differences when processing medical data. Through systematic comparison, it is possible to identify the model most suitable for cardiovascular disease prediction tasks while revealing the dataset's intrinsic characteristics. The selection of evaluation metrics followed the specific requirements of medical applications, and four metrics were adopted in this study: accuracy, F1 score, accuracy, and recall. The overall prediction accuracy of the accuracy measurement model is the most intuitive performance metric; F1 score, as a reconciled average of accuracy and recall, can provide a more comprehensive performance assessment in the case of category imbalances; accuracy focuses on the proportion of samples predicted to be positive that are actually positive, corresponding to the control of false positives in medical diagnosis; and recall focuses on the proportion of samples predicted correctly that are actually positive, corresponding to the control of false negatives in medical screening, which is important for avoiding patients at high risk of misdiagnosis.

V. Model Development and Training

The model development phase employs a standardized machine learning process to ensure the scientificity of the experiments and the reproducibility of the results.

First, the pre-processed dataset is divided into the training and test sets at an 80:20 ratio, with the training set containing 54,989 samples and the test set containing 13,748 samples. The division process employs hierarchical sampling strategies to ensure that the distribution of target variables in the training and test sets is consistent with the original dataset. Second, for models sensitive to feature scale, such as logical regression and K-nearest neighbors, standardized pre-processing is applied to convert the features to a normal distribution with an mean of 0 and a standard deviation of 1. For tree-based models, such as decision trees, random forests, and gradient enhancement trees, due to their inherent invariance, the raw features are used directly for training.

We used the grid search, a strategy that combines pentagonal cross-validation, to systematically explore the grid search space while evaluating the generalization performance of the model through cross-validation to avoid overfitting. For logistic regression, the normalized parameter C value of the search is [0.1, 1.0, 10.0], with a fixed maximum iteration count of 1000; for K-nearest neighbor classifiers, the number of neighbors of the search is [5, 10, 15], with a weighted strategy of ['uniform', 'distance']; for decision trees, the maximum depth of the search is [5, 10, 15], with a minimum split sample count of [10, 50, 100]; for random forests, the number of trees of the search is [50, 100], with a maximum depth of [10, 15, 20]; for gradient boost trees, the number of trees of the Search is [50, 100], the learning rate is [0.05, 0.1], with a maximum depth of [3, 5]. The grid search process evaluates the performance of each set of hyperparameter combinations under a framework of fifty-fifty cross-validation with accuracy as an optimized goal.

The experimental results (Table 1, Figure 3) showed that the gradient enhancement tree model achieved optimal performance on all evaluation metrics with an accuracy of 0.7321, F1 score of 0.7182, accuracy of 0.7490, recall rate of 0.6898, and AUC value of 0.7991. The decision tree model followed closely with an accuracy of 0.7315, its accuracy being the highest among all models, reaching 0.7580, indicating that this model has high reliability in predicting patients with cardiovascular disease. The performance of the random forest model is slightly lower than that of the decision tree, with an accuracy of 0.7289, a result that may stem from the introduction of randomness in the random forest that reduces the model's determinism to some extent. Logical regression as a representative of linear models, with an accuracy of 0.7253, indicates that there is a certain linear relationship between features and target variables, but nonlinear models are able to capture more complex patterns. The accuracy of the K-nearest neighbor classifier is 0.7167, the lowest among all models, which may be because the distance measure in high-dimensional feature space is affected by the dimension curse.

We performed a feature importance analysis based on a random forest model (Table 2), revealing the extent to which each feature contributes to cardiovascular disease prediction. systolic blood pressure (ap_hi) ranked first with an importance score of 0.318, indicating that blood pressure levels are the most critical factor for cardiovascular disease prediction, a finding that is highly consistent with medical common sense. Age ranked second with an importance score of 0.174, reflecting the epidemiological pattern of increased cardiovascular disease risk with age. Diastolic blood pressure (ap_lo) ranked second with an importance of 0.138, which together with systolic blood pressure constitutes the overall contribution of blood pressure factors. The importance of the body mass index (BMI) was 0.104, highlighting the importance of obesity as a risk factor for cardiovascular disease. The importance of body weight and height was 0.077 and 0.067, respectively, the importance of cholesterol levels was 0.066, and the importance of blood sugar levels was 0.017. Notably, the importance of lifestyle factors such as physical activity, smoking, and alcohol consumption was relatively low, being 0.012, 0.009, and 0.007, respectively, which may be because these factors indirectly affect cardiovascular disease risk by influencing physiological indicators such as blood pressure and body weight.

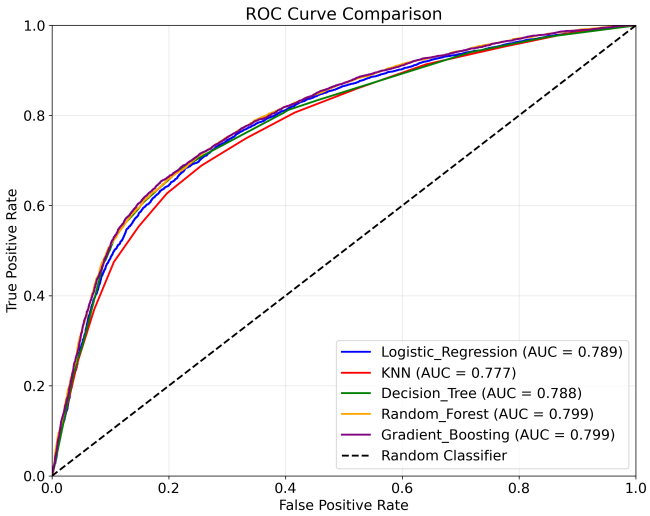

Fig. 5. ROC curve comparison of models.

Confusion matrix analysis further reveals the predictive characteristics of each model (Figure 5). The gradient-boosting tree model achieved a better balance in the identification of true and true-negative cases, with a relatively balanced number of false positives and false negatives. The decision tree model, although with the highest accuracy, had a relatively low recall rate, indicating that it tended to make conservative predictions, reducing false positives but increasing false negatives. This characteristic needs to be weighed in the medical screening scenario, as false negatives mean missing high-risk patients, which can lead

TABLE I
Model comparison results

Model	Accuracy	F1_Score	Precision	Recall	AUC
Gradient_Boosting	0.7321	0.7182	0.7490	0.6898	0.7991
Decision_Tree	0.7315	0.7124	0.7580	0.6719	0.7875
Random_Forest	0.7289	0.7121	0.7504	0.6775	0.7987
Logistic_Regression	0.7253	0.7058	0.7510	0.6657	0.7893
KNN	0.7167	0.7065	0.7249	0.6890	0.7775

TABLE II
Feature importance ranking.

Feature	Importance
ap_hi	0.3176
age	0.1744
ap_lo	0.1378
bmi	0.1040
weight	0.0773
height	0.0669
cholesterol	0.0655
gluc	0.0174
active	0.0123
gender	0.0107
smoke	0.0087
alco	0.0075

to serious consequences. The confusion matrix of the logistic regression model showed a similar pattern, with a slightly higher number of false negatives than false positives. The K-nearest neighbors model had a relatively close number of false positives and false negatives, but a slightly lower overall accuracy rate.

VI. Conclusions

Combining the above analyses, this study successfully developed a system of machine learning models for cardiovascular disease risk prediction, providing data-driven solutions for early screening and risk assessment through systematic model comparison and in-depth feature analysis. The gradient enhancement tree model became the preferred model for its superior comprehensive performance, with systolic pressure, age, and diastolic pressure being identified as the most important predictive features. These findings not only validate known medical knowledge but also provide a quantitative basis for risk factor interventions in clinical practice. Future research can further explore deep learning methods, feature engineering techniques, and model explainability enhancements to drive the application of cardiovascular disease prediction systems in real-world medical scenarios [2], [9], [10].

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